GRADE 7 ACCELERATED · UNIT 5 · ASSESSMENT

Multi-Step Proportional Relationships

Form B · 24 points **WORKED ANSWERS**

Show your work. One point for each lettered part.

Part 1. Scale Drawings and Scale Factors *10 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

1. A 6 cm side of a figure is 9 cm in its scale copy. Find the scale factor. *1 pt*

k = 1.5

copy ÷ original

Scoring: 1 point.

2. A 20 cm side of a figure is 5 cm in its scale copy. Find the scale factor. *1 pt*

k = 0.25

copy ÷ original; the copy is smaller, so k is under 1 — 4 is the proportion backwards

Scoring: 1 point.

3. A figure is scaled by a scale factor of 4. Select ALL of the statements that are true about the copy. *1 pt*

- ☐ A. Each angle is 4 times as large.
- ☒ **B. The area is 16 times as large.**
- ☒ **C. Each side is 4 times as long.**
- ☐ D. The area is 4 times as large.
- ☒ **E. The perimeter is 4 times as long.**

B, C and E

All three and nothing else for the point: sides and perimeter × k, area × k^2, angles unchanged.

Scoring: 1 point.

4. A 2 cm by 5 cm rectangle is scaled by k = 4.

a. Find the copy's perimeter. *1 pt*

56 cm

perimeter × k

b. Find the copy's area. *1 pt*

160 cm²

area × k² — 40 multiplied the area by k once

Scoring: a 1 · b 1 = 2 points.

5. A figure has area 5 cm². Its scale copy has area 80 cm². What is the scale factor? *1 pt*

k = 4

80 ÷ 5 = 16 is k², so k = √16. A board with 16 handed in the area ratio as k.

Scoring: 1 point.

6. A floor plan is drawn at ½ in : 1 ft. A room is 3 in by 5 in on the plan.

a. Find the real length of the wall drawn 3 in long. *1 pt*

6 ft

the scale per 1 inch, then ×: 1 in : 2 ft

b. Find the room's real area. *1 pt*

60 ft²

real dimensions first: 6 by 10 — 15 in² × 2 = 30 is the named error

Scoring: a 1 · b 1 = 2 points.

7. A plan is drawn at ¼ in : 1 ft. A real wall is 22 ft long. How long is it on the plan? *1 pt*

5.5 in

real ÷ rate

Scoring: 1 point.

8. A plan is drawn at 1 in : 4 ft. A room is 1.5 in by 2 in on the plan. 1 ft = 0.305 m. Find the room's real area in square meters. *1 pt*

About 4.47 m² (1.83 m by 2.44 m); 4.46 by the sheet's other ratio

6 ft by 8 ft; 1.83 × 2.44 = 4.4652. The same number by the other route: 48 ft² × 0.305² = 4.4652; by the sheet's 1 m = 3.28 ft, 48 ÷ 3.28² = 4.46 — full credit. 14.64 (48 × 0.305) used the ratio once on an area. A plan's area asked in metric appears on no review and in no question bank; its type was 5.10 board 8.

Scoring: 1 point.

Part 2. Converting Across Systems

8 points

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

9. 1 in = 2.54 cm. How many centimeters is 11 in? *1 pt*

27.94 cm (28.2 by ÷ 0.39)

× the ratio whose 1 is inches; the sheet's other ratio, ÷ 0.39, earns the point too when the work shows it

Scoring: 1 point.

10. 1 in = 2.54 cm and 1 cm = 0.39 in. Select ALL of the expressions that give the number of inches in 90 cm. *1 pt*

- ☐ A. 90 ÷ 0.39
- ☒ **B. 90 ÷ 2.54**
- ☐ C. 90 × 2.54
- ☒ **D. 90 × 0.39**

B and D

Both sheet routes (35.1 and 35.4) and nothing else for the point.

Scoring: 1 point.

11. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimeters is 5 yd? *1 pt*

457.2 cm

a chain: 180 in, then × 2.54 (180 ÷ 0.39 = 461.5 by the other ratio — full credit). The estimate catches 12.7.

Scoring: 1 point.

12. 1 lb = 0.454 kg. How many kilograms is 60 lb? *1 pt*

27.24 kg (27.22 by ÷ 2.204)

× the ratio whose 1 is the unit you have; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

13. 1 gal = 3.785 L. How many liters is 9 gal? *1 pt*

34.065 L (34.09 by ÷ 0.264)

more liters than gallons; the sheet's other ratio earns the point too when the work shows it

Scoring: 1 point.

14. 1 m = 3.28 ft. A floor is 5 m^2 in area. Which is closest to its area in square feet? *1 pt*

- ● **A. 53.8 ft^2**
- ○ B. 16.4 ft^2
- ○ C. 10.8 ft^2
- ○ D. 32.8 ft^2

A

5 × 3.28^2 = 53.79 (or 5 ÷ 0.305^2 = 53.75).

Scoring: 1 point.

15. Rate on Nov 5: 1 US dollar = 1.35 Canadian dollars.

a. How many Canadian dollars is $70? *1 pt*

94.50 Canadian dollars

× the rate

b. How many US dollars is 54 Canadian dollars? *1 pt*

$40

÷ the rate; 72.90 multiplied both ways

Scoring: a 1 · b 1 = 2 points.

Part 3. Multi-Step Proportion Problems *6 points*

If an earlier part is wrong, a later part still earns its point when the right operation is applied to your own earlier answer — but only if that work is on the page.

16. A car gets 22 miles per gallon and its tank holds 20 gal.

a. How far does it go on a full tank? *1 pt*

440 mi

the rate × the gallons

b. It makes a 1,500-mi trip. How many tanks of gas does the trip use? (Part of a tank counts as a whole tank.) *1 pt*

4 tanks (1,500 ÷ 440 ≈ 3.41)

the bracket on the paper settles the rounding; follow-through on part (a)

Scoring: a 1 · b 1 = 2 points.

17. A recipe for 6 servings uses 2 cups of rice.

a. How much rice for 9 servings? *1 pt*

3 cups

k = 1.5 from the yields, then the ingredient

b. With 8 cups on hand, how many servings can be made? *1 pt*

24 servings

4 recipes × the yield; 48 scaled the yield by the cups

Scoring: a 1 · b 1 = 2 points.

18. Paint is mixed 1 : 3 : 4. A batch uses 12 gal of the second color. How many gallons is the whole batch? *1 pt*

32 gal

k = 4 from the second color, then the whole, 8 parts; 16 found the third color and stopped

Scoring: 1 point.

19. 1 mi = 1.61 km. A drive is 450 km long. Truck A averages 55 mi/h; truck B averages 90 km/h. Which truck gets there first, and by about how many minutes? Show both speeds in one unit. *1 pt*

Truck B, by about 5 minutes: 55 mi/h = 88.55 km/h, so 450 ÷ 90 = 5 h against 450 ÷ 88.55 ≈ 5.08 h.

The point needs the speeds in one unit and the choice with its gap: 0.082 h is 4.9 min — 5 earns it (by the 0.62 route, 279 mi: 5 h against 5.07 h, 4 min; accept 4–5). 'Truck B, 90 > 55' is right by accident — no credit without the conversion. The surface appears on no review and in no question bank; its type was 5.10 board 9.

Scoring: 1 point.